LectureAssist: An Agentic, Multimodal AI Assistant for Lecture-Grounded Study Material

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Problem

- Lecture knowledge is locked in **video, speech and board writing**.
- Generic quiz tools produce **ungrounded, off-topic** questions.
- Fixed difficulty ignores **each student's weak topics**.

Goal. Grounded, validated, adaptive study material — fully local, **zero API cost**, on one free Colab T4.

Evaluation Setup

- **Source.** OpenStax *Calculus Vol. 1*, Limits chapter (≈112k characters), also given to the judge as ground truth.
- **Scale.** n=200 MCQ per pipeline — easy 68, medium 66, hard 66.
- **Fairness.** Identical sliding window for both pipelines, so any difference reflects **prompting strategy alone**.
- **Models.** Generator `gemma3:4b`, judge `gemma3:12b`, both local.

The System in Use

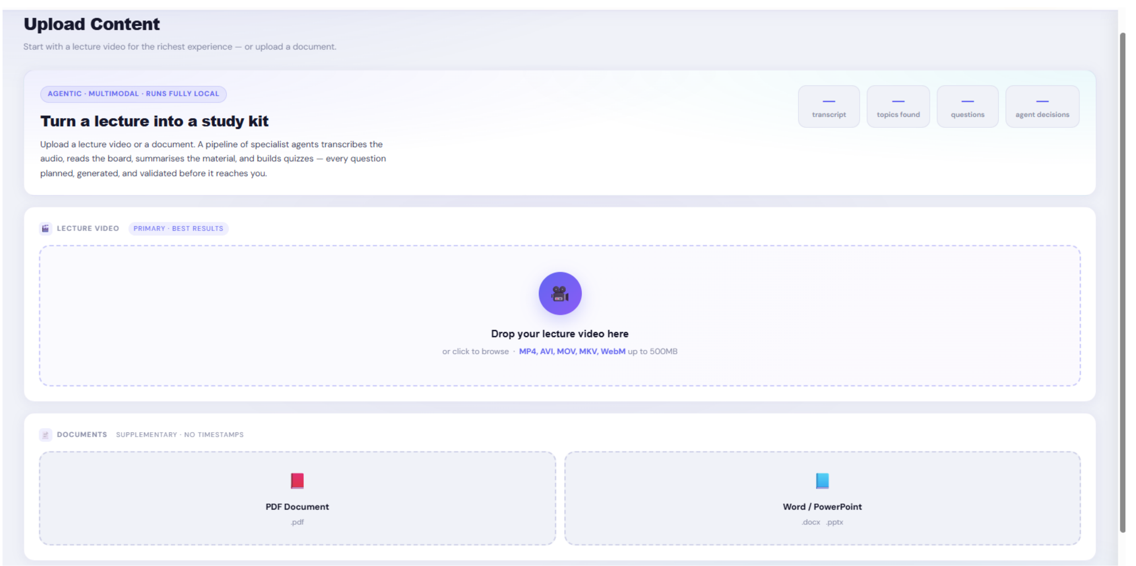

Figure 1. Upload page. Drag-and-drop for video or documents; the four counters fill in from the run once processing completes.

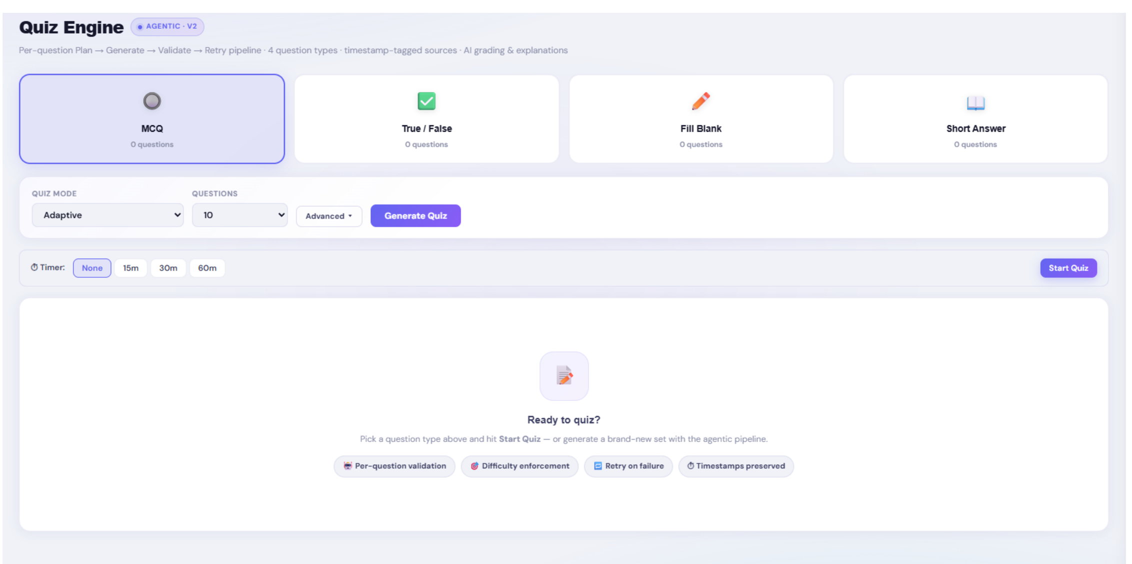

Figure 2. Quiz Engine. Four question types, generation modes, and a timer; the agent workspace streams the pipeline live during generation.

System Architecture

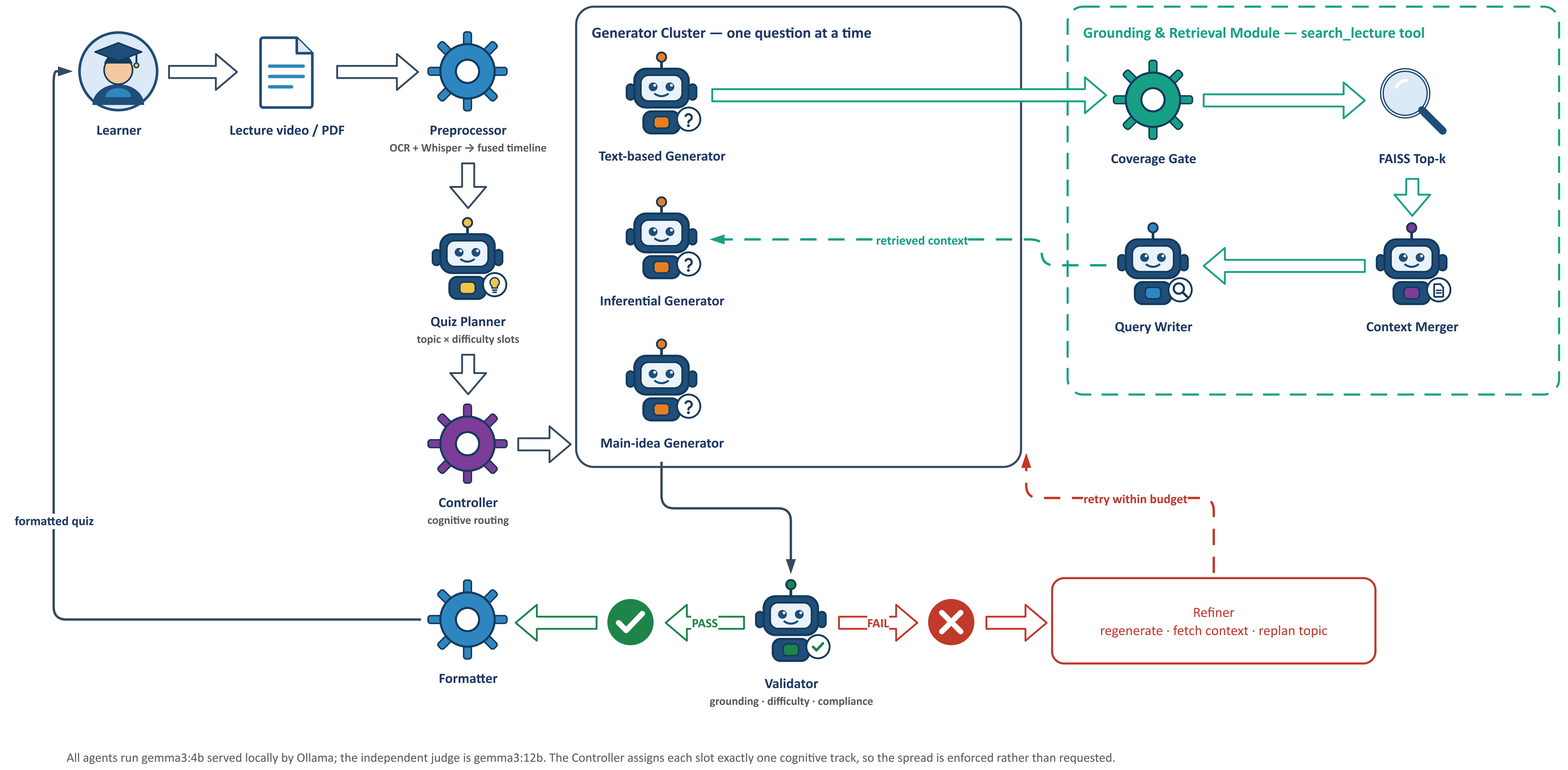

Figure 3. Multimodal extraction feeds a grounded knowledge base; the orchestrator coordinates the specialist agents that produce every study artifact. Independent agents run concurrently and any agent crash is isolated.

Agentic Quiz Generation

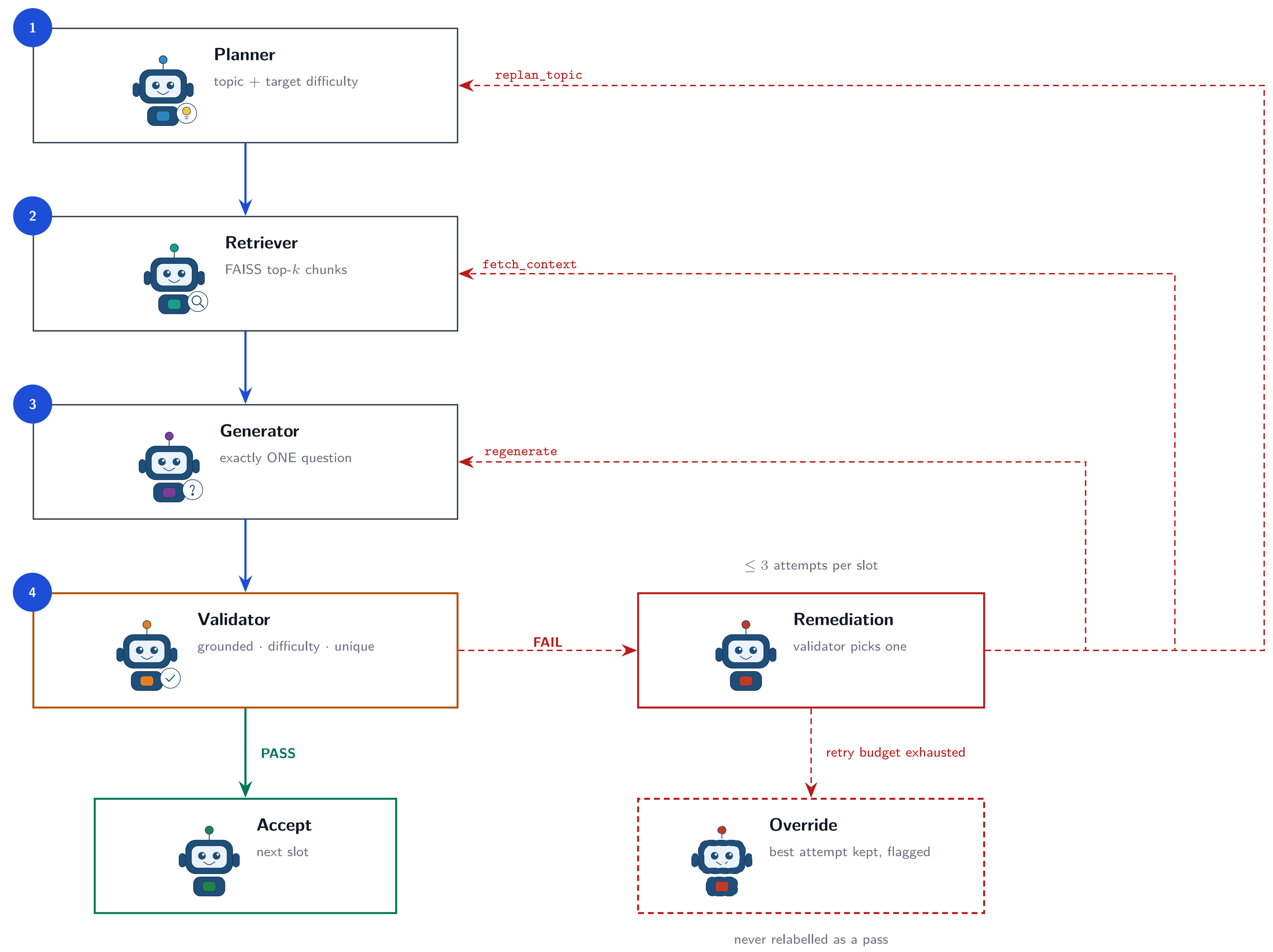

Figure 4. One question per slot; each must pass validation before the next begins. An exhausted retry budget flags an override, never a pass.

Results

Metric	Agen.	Non-ag.
Judged 1–5 (saturated)		
Overall	4.86	4.93
Correctness	4.98	5.00
Clarity	4.92	4.97
Difficulty	4.36	4.39
Automated 0–1		
Duplicate ↓	0.385	0.280
Grounding	0.557	0.624
Eff. accept	0.612	0.720
Answer match	0.940	0.970
Similarity ↓	0.932	0.894
Accept rate	0.995	1.000
Verified	0.995	0.995

Diff.	Pipe.	Dup.↓	Grnd.	Eff.	Match
Easy	Agen.	0.353	0.562	0.647	0.971
	Non-ag.	0.324	0.642	0.676	0.971
Medium	Agen.	0.379	0.595	0.612	0.879
	Non-ag.	0.227	0.614	0.773	0.955
Hard	Agen.	0.288	0.513	0.712	0.970
	Non-ag.	0.182	0.616	0.818	0.985

The pattern holds at every difficulty: the baseline leads on duplicate rate, grounding and effective acceptance in all three bands.

n=200 per pipeline. Green marks the better value; ↓ marks rows where lower is better. Per-difficulty n=68/66/66.

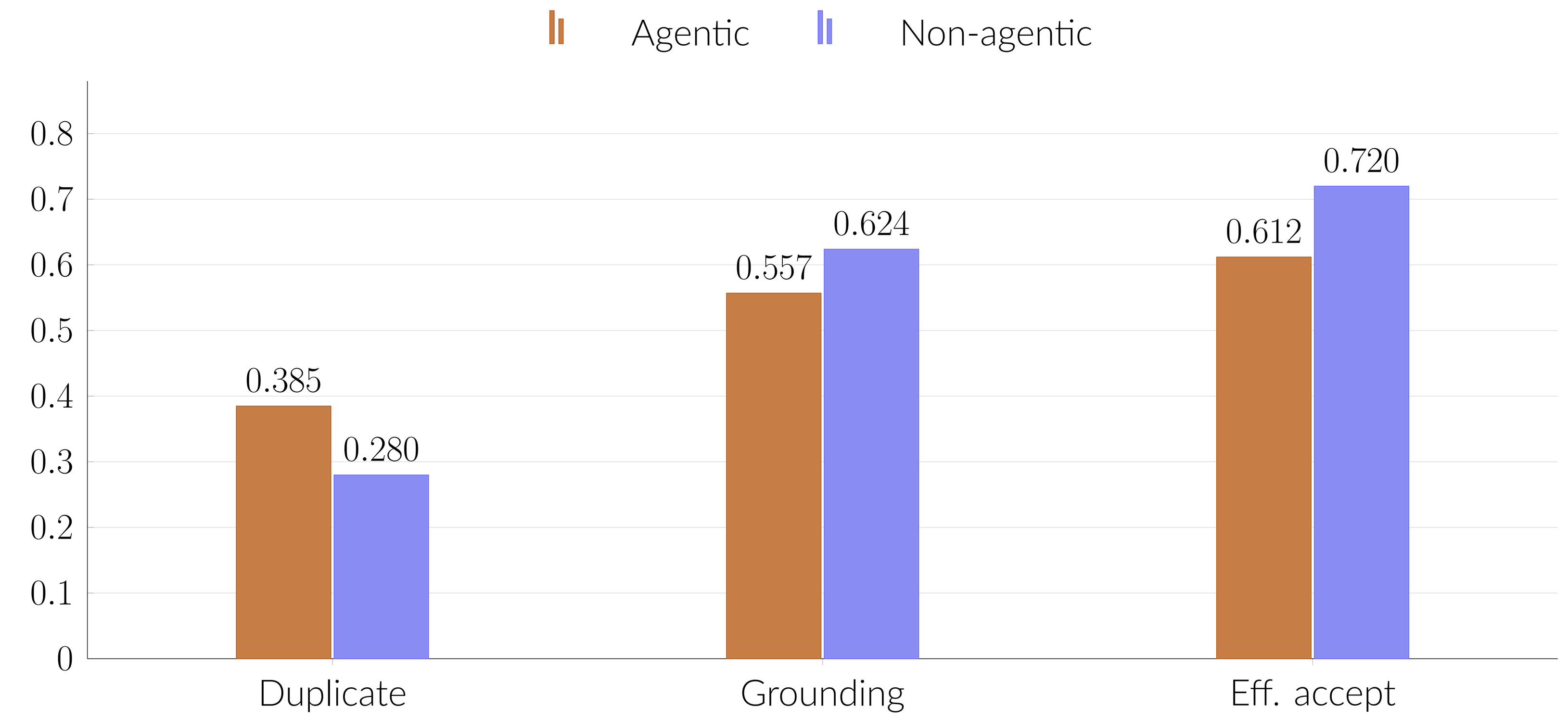

Figure 5. The three metrics that discriminate. Lower is better for duplicate rate, so the baseline leads on all three.

Conclusion

A fully local, $0-API multimodal pipeline that turns a lecture into grounded, validated study material. The judged dimensions **saturate** (4.36–5.00), so only the automated metrics discriminate — and on those the agentic loop does **not** beat single-pass prompting. The engineering contribution stands; the scientific claim is **not established**, and we report it that way.

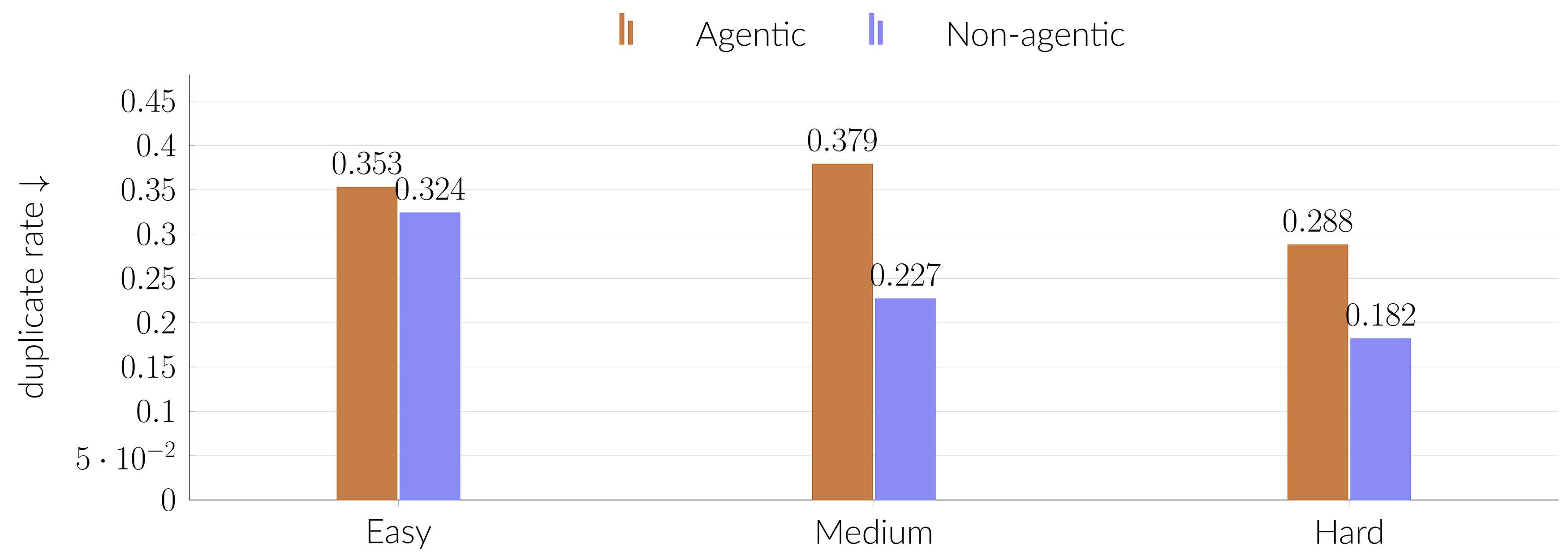

Figure 6. Duplicate rate by difficulty — the agentic loop duplicates more in every band.